Demonstration of Proposed Features

OptiCrop: Smart Agricultural Production Optimization Engine

Date	28 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S. No	Feature Name	Description	Status	Demonstrated (Yes / No)	Remarks
1	Crop Recommendation System	Takes N, P, K, temperature, humidity, pH, and rainfall as input and recommends the most suitable crop using ML classification.	Implemented	Yes	Random Forest model with 99% accuracy; tested with live inputs
2	Crop Suitability Assessment	Evaluates whether current soil and climate conditions are suitable for a user-specified crop and provides a compatibility score.	Implemented	Yes	Integrated with Flask UI; results shown with suitability percentage
3	Data Visualization Dashboard	Visual representation of crop-environment relationships using Matplotlib and Seaborn — includes heatmaps, bar charts, and scatter plots.	Implemented	Yes	Charts embedded in the web app and Jupyter Notebook
4	Flask Web Application	User-friendly web interface built with Flask for farmers and researchers to input parameters and receive recommendations interactively.	Implemented	Yes	Responsive UI; deployed locally on Flask dev server
5	Model Comparison & Evaluation	Comparison of multiple ML models (Decision Tree, Random Forest, Naive Bayes, SVM) with accuracy, precision, recall, and F1-score metrics.	Implemented	Yes	Random Forest selected as best model; all metrics documented

S. No	Feature Name	Description	Status	Demonstrated (Yes / No)	Remarks
6	Agricultural Research Analytics Module	Allows researchers/policymakers to analyze patterns in crop-environment data to support evidence-based agricultural planning.	Partial	Yes	Core analytics implemented; advanced export features deferred to future scope

Feature Implementation Summary

Total Features Proposed	6
Total Features Implemented	5
Total Features Demonstrated	6
Overall Implementation Rate (%)	91.67% (5 fully implemented, 1 partial)